

CALCULUS WTW158
EXAMINATION / EKSAMEN

Date / Datum: June / Junie 2017 Marks / Punte : 55 Time / Tyd : 3 hours / uur

M NUMBER / M-NOMMER

MARKS FOR SECTIONS B & C
PUNTE VIR AFDELINGS B & C

MEMO

40

SURNAME / VAN

FIRST NAMES / VOORNAME _____

STUDENT NUMBER

STUDENTENOMMER

SIGNATURE

HANDTEKENING

CONTACT NUMBER

KONTAKNOMMER

Internal Examiners / Interne Eksaminatore

WTW 158 Lecturers

External Examiner / Eksterne Eksaminator

Dr D Moubandjo

READ THE FOLLOWING INSTRUCTIONS

1. The paper consists of pages 1 to 13. Check whether your paper is complete.
2. No calculator may be used.
3. No question paper may be removed from the venue.

LEES DIE VOLGENDE INSTRUKSIES

1. Die vraestel bestaan uit bladsye 1 tot 13. Kontroleer of u vraestel volledig is.
2. Geen sakrekenaar mag gebruik word nie.
3. Geen vraestel mag uit die lokaal geneem word nie.

Section A: / Afdeling A:

Marks: 15 / Punte: 15

1. The questions in Section A must be completed on SIDE 2 of the optical reader form in SOFT PENCIL.
 2. First circle your answers on this paper and then transfer these to the optical reader form.
 3. There is only ONE correct answer per question.
-
1. Die vrae in Afdeling A moet op KANT 2 van die merkleesvorm in SAGTE POTLOOD voltooi word.
 2. Omkring eers jou antwoorde op hierdie vraestel en dra dit dan oor na die merkleesvorm.
 3. Daar is slegs EEN korrekte antwoord per vraag.

Question 1 / Vraag 1

The solution of $\frac{|x-5|}{x+1} > 0$ is

Die oplossing van $\frac{|x-5|}{x+1} > 0$ is

(a)	$x < -1$	(b)	$x < -1$ or/of $x > 5$	(c)	$-1 < x < 5$
(d)	$x > -1$ and/en $x \neq 5$	(e)	$x > 5$	(f)	None of these / Geen van hierdie

Question 2 / Vraag 2

$\lim_{t \rightarrow 1^-} \frac{4}{t^2 - 1} =$

(a)	0	(b)	$-\infty$	(c)	∞
(d)	-1	(e)	None of these / Geen van hierdie		

Question 3 / Vraag 3

The inverse function of $f(x) = e^x$, $x \in [0, 1]$ is

Die inverse funksie van $f(x) = e^x$, $x \in [0, 1]$ is

(a)	$y = e^{-x}, x \in [0, 1]$	(b)	$y = \ln x, x \in [0, 1]$	(c)	$y = e^{\frac{1}{x}}, x \in [1, e]$
(d)	$y = \ln x, x \in [1, e]$	(e)	None of these / Geen van hierdie		

Question 4 / Vraag 4

For the function $f(x) = \frac{2x+1}{2x^2-x-1}$, which one of the following is true?

Vir die funksie $f(x) = \frac{2x+1}{2x^2-x-1}$, watter een van die volgende is waar?

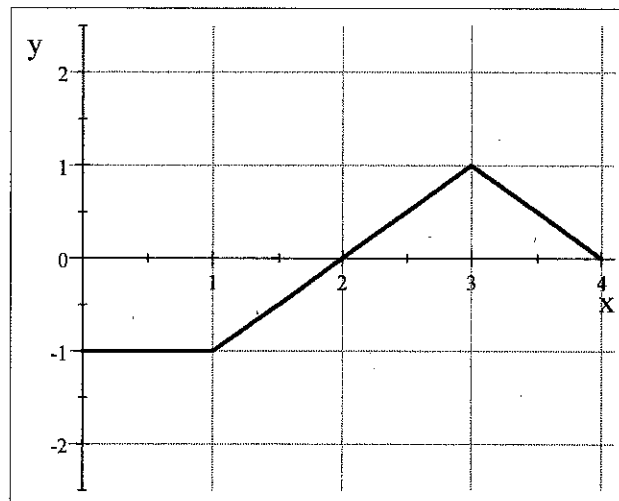
(a)	f has a horizontal asymptote $y = 0$. / f het 'n horisontale asimptoot $y = 0$.
(b)	f has a horizontal asymptote $y = 1$. / f het 'n horisontale asimptoot $y = 1$.
(c)	f has a horizontal asymptote $y = -\frac{1}{2}$. / f het 'n horisontale asimptoot $y = -\frac{1}{2}$.
(d)	f has no horizontal asymptotes. / f het geen horisontale asimptote nie.

The following information applies to Questions 5 and 6.

Die volgende inligting het betrekking op Vrae 5 en 6.

The graph of a piecewise defined function f on $0 \leq x \leq 4$ is given:

Die grafiek van 'n stuksgewys gedefinieerde funksie f op $0 \leq x \leq 4$ word gegee:



Question 5 / Vraag 5

$$\int_0^4 f(x) dx =$$

(a)	-0.5	(b)	-1.5	(c)	2.5	(d)	0.5
(e)	3	(f)	None of these / Geen van hierdie				

Question 6 / Vraag 6

The area enclosed between the curve and the x -axis on $[0, 4]$ is given by

Die oppervlakte ingesluit tussen die kromme en die x -as op $[0, 4]$ word gegee deur

(a)	-0.5	(b)	-1.5	(c)	2.5	(d)	0.5
(e)	3	(f)	None of these / Geen van hierdie				

Question 7 / Vraag 7

The Riemann sum approximation of $\int_0^4 x^2 dx$, using the left-hand sum with four subintervals, is
Die Riemann-som benadering van $\int_0^4 x^2 dx$, deur die linkersom te gebruik met vier sub-intervalle, is

(a)	30	(b)	14	(c)	10
(d)	25	(e)	None of these / Geen van hierdie		

Question 8 / Vraag 8

Consider a function f with the following properties: / Beskou 'n funksie f met die volgende eienskappe:

- f is continuous and differentiable on $[0, 5]$. / f is kontinu en differensieerbaar op $[0, 5]$.
- f has critical numbers $x = 2$ and $x = 4$. / f het kritieke getalle $x = 2$ en $x = 4$.
- The following table gives function values of f : / Die volgende tabel gee funksiewaardes van f :

x	0	2	4	5
$f(x)$	2	1	2	6

Select the correct statement. / Kies die korrekte bewering.

The function has / Die funksie het

(a)	a local maximum at $x = 2$ and a inflection point at $x = 4$.
(b)	a local minimum at $x = 2$ and a local maximum at $x = 4$.
(c)	a local minimum at $x = 2$ and an inflection point at $x = 4$.
(d)	local minima at both $x = 2$ and $x = 4$.
(e)	Too little information is given to determine the extremes.

(a)	'n lokale maksimum by $x = 2$ en 'n infleksiepunt in $x = 4$.
(b)	'n lokale minimum in $x = 2$ en 'n lokale maksimum in $x = 4$.
(c)	'n lokale minimum in $x = 2$ en 'n infleksiepunt in $x = 4$
(d)	lokale minima in beide $x = 2$ en $x = 4$.
(e)	Te min inligting word gegee om die ekstreme te bepaal.

Question 9 / Vraag 9

The distance from the point $(2, 1, 3)$ to the xy -plane is

Die afstand van die punt $(2, 1, 3)$ na die xy -vlak is

(a)	3	(b)	$\sqrt{13}$	(c)	$\sqrt{5}$	(d)	2
(e)	None of these / Geen van hierdie						

Question 10 / Vraag 10

Given that $|\vec{a}| = 3$, $|\vec{b}| = \sqrt{6}$ and the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$. Determine $\vec{a} \cdot \vec{b}$.

Gegee dat $|\vec{a}| = 3$, $|\vec{b}| = \sqrt{6}$ en die hoek tussen $\vec{a}$ en $\vec{b}$ is $\frac{\pi}{4}$. Bepaal $\vec{a} \cdot \vec{b}$.

(a)	$\frac{3\sqrt{6}}{2}$	(b)	$\frac{9\sqrt{3}}{2}$	(c)	$3\sqrt{3}$	(d)	$\frac{1}{6\sqrt{3}}$
(e)	$\frac{2\sqrt{3}}{9}$	(f)	$\frac{1}{3\sqrt{6}}$	(g)	None of these / Geen van hierdie		

Afdeling B: / Section B:

Marks: 12 / Punte: 12

This section must be completed IN PEN in this paper.

Fill in the answer only in the space provided. Rough work can be done on the facing page and will not be marked.

Hierdie afdeling moet IN PEN op hierdie vraestel gedoen word.

Vul slegs die antwoord in die spasie in. Rofwerk kan op die teenblad gedoen word en sal nie gemerk word nie.

Question 1 / Vraag 1

Find the domain of $f(x) = \frac{\ln(x+1)}{(x-1)^2}$.

Vind die definisieversameling van $f(x) = \frac{\ln(x+1)}{(x-1)^2}$.

$$x \in (-1, \infty) \setminus \{1\} \quad \text{or} \quad (-1, 1) \cup (1, \infty)$$

or $x > -1, x \neq 1$

Question 2 / Vraag 2

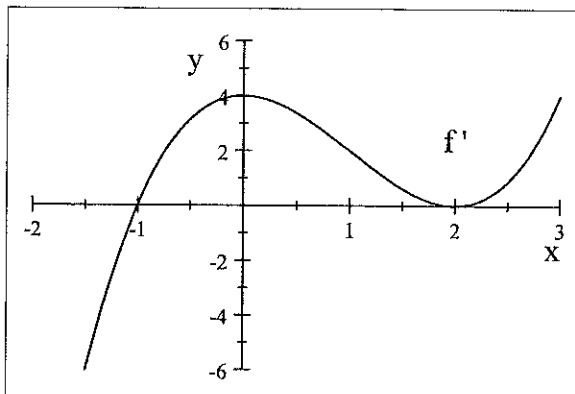
$\lim_{x \rightarrow \infty} \arctan x = \lim_{x \rightarrow \infty} \tan^{-1} x =$

$$\frac{\pi}{2}$$

Question 3 / Vraag 3

The graph of the **derivative function** f' of a function f is shown, with local extremes at $x = 0$ and $x = 2$, and with $D_f = \mathbb{R}$.

Die grafiek van die **afgeleide funksie** f' van 'n funksie f word gegee met lokale ekstreme by $x = 0$ en $x = 2$, en met $D_f = \mathbb{R}$.



3.1 Give the interval(s) on which **the function** f is decreasing.

Gee die interval(le) waarop **die funksie** f daal.

$(-\infty, -1)$

3.2 Identify the local extremes of **the function** f .

Identifiseer die lokale ekstreme van **die funksie** f .

local minimum in $x = -1$

(inflection point in $x = 2$, not an extreme)

not for marks

3.3 Identify the interval(s) on which **the function** f is concave down.

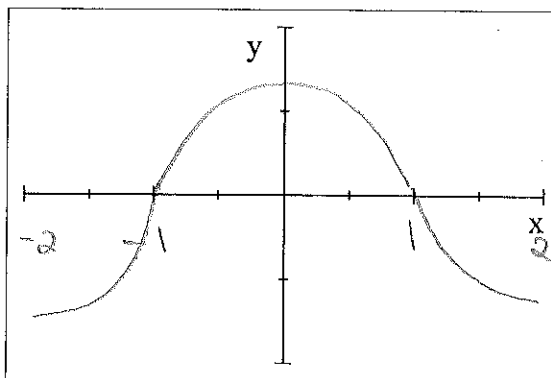
Identifiseer die interval(le) waar **die funksie** f konkaf na onder is.

$(0, 2)$

Question 4 / Vraag 4

Sketch a differentiable function f on $[-2, 2]$ such that $f'' < 0$ for $|x| < 1$ and $f'' > 0$ for $|x| > 1$.

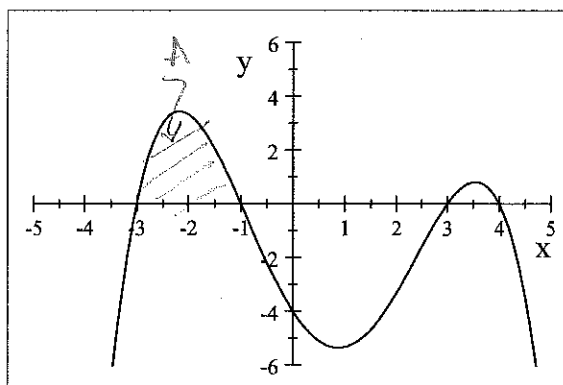
Skets 'n differensieerbare funksie f op $[-2, 2]$ sodat $f'' < 0$ vir $|x| < 1$ en $f'' > 0$ vir $|x| > 1$.



Shape important
Function can
move up or
down

Question 5 / Vraag 5

Consider the graph of a function f :
Beskou die grafiek van 'n funksie f :



$$A - 5 + 1.5 = 0.5$$

$$\Rightarrow A = 4$$

Given that / Gegee dat $\int_{-3}^4 f(x) dx = 0.5$, $\int_{-1}^3 f(x) dx = -5$ and / en $\int_3^4 f(x) dx = 1.5$.

The area of the region enclosed between the graph and the x -axis on $[-3, 4]$ is

Die oppervlakte van die gebied ingesluit tussen die grafiek en die x -as op $[-3, 4]$ is

$$10.5$$

1

Question 6 / Vraag 6

$$\int_0^{\pi} \sin x dx =$$

$$2$$

1

Question 7 / Vraag 7

$$\int_{-1}^2 |2x| dx =$$

$$5$$

1

Question 8 / Vraag 8

A vector orthogonal to both $\vec{a} = \langle 2, 0, -1 \rangle$ and $\vec{b} = \langle 1, 4, 3 \rangle$ is

'n Vektor ortogonaal op beide $\vec{a} = \langle 2, 0, -1 \rangle$ en $\vec{b} = \langle 1, 4, 3 \rangle$ is

$$\langle 4, -7, 8 \rangle \text{ or } \langle -4, 7, -8 \rangle$$

1

Question 9 / Vraag 9

Find a vector of length 4 in the opposite direction to $\langle 1, -2, 2 \rangle$.

Bepaal 'n vektor van lengte 4 in die teenoorgestelde rigting as $\langle 1, -2, 2 \rangle$.

$$\left\langle -\frac{4}{3}, \frac{8}{3}, -\frac{8}{3} \right\rangle$$

1

Afdeling C: / Section C:

Marks: 28 / Punte: 28

1. The question in Section C must be completed in pen on this paper.
2. All calculations must be shown and will be marked.
3. Attention should be paid to writing in a mathematically sound way.

1. Die vrae in Afdeling C moet op die vraestel in ink voltooi word.
2. Alle berekeninge moet getoon word en sal gemerk word.
3. Maak seker dat jou skryfwyse wiskundig korrek is.

Question 1 / Vraag 1

Determine $\frac{dy}{dx}$ in each of the following cases. / Bepaal $\frac{dy}{dx}$ in elk van die volgende gevalle.

Do not simplify your answers. / Moenie jou antwoorde vereenvoudig nie.

1.1 $y = e^{x^2 \cos \sqrt{x}}$

$$\frac{dy}{dx} = e^{x^2 \cos \sqrt{x}} \left(2x \cos \sqrt{x} - x^2 \cdot \sin \sqrt{x} \cdot \frac{1}{2\sqrt{x}} \right)$$

1.2 $y = \frac{3x^2 + 7}{\cot(3x)}$

$$\frac{dy}{dx} = \frac{6x \cdot \cot 3x + (3x^2 + 7) \operatorname{cosec}^2(3x) \cdot 3}{(\cot(3x))^2}$$

1.3 $x^3 + y = y^3$

$$3x^2 + \frac{dy}{dx} = 3y^2 \cdot \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{3x^2}{3y^2 - 1}$$

Question 2 / Vraag 2

Evaluate the following integrals. / Bepaal die volgende integrale.

$$2.1 \int \frac{x}{\sqrt{1-x^2}} dx = -(1-x^2)^{1/2} + C \quad 1$$

$$2.2 \int \frac{x}{\sqrt{1-x^4}} dx = \frac{1}{2} \arcsin(x^2) + C \quad 1$$

$$2.3 \int \cot 3x dx = \int \frac{\cos 3x}{\sin 3x} dx = \frac{1}{3} \ln |\sin 3x| + C \quad 2$$

$$2.4 \int \frac{\sec^2 \sqrt{x}}{\sqrt{x}} dx = 2 \tan \sqrt{x} + C \quad 2$$

Question 3 / Vraag 3

Determine the limit, if it exists. / Bepaal die limiet, indien dit bestaan.

$$\lim_{x \rightarrow 0^+} (1+2x)^{3/x}$$

$$\begin{aligned} \text{Let } y &= (1+2x)^{3/x} \\ \Rightarrow \ln y &= \frac{3}{x} \ln(1+2x) \\ \Rightarrow \lim_{x \rightarrow 0^+} \ln y &= \lim_{x \rightarrow 0^+} \frac{3 \ln(1+2x)}{x} \quad \left(\frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0^+} \frac{\frac{6}{1+2x}}{1} \\ &= 6 \end{aligned} \quad 3$$

$$\Rightarrow \lim_{x \rightarrow 0^+} y = e^6$$

Question 4 / Vraag 4

4.1 Complete the definition: A function f is increasing on an interval I if

Voltooi die definisie: 'n Funksie f is stygend op 'n interval I as

$f(x_2) > f(x_1)$ for any $x_1, x_2 \in I$
such that $x_1 < x_2$

1

4.2 Prove that if $f'(x) > 0$ on an interval I , then f is increasing on the interval I .

Bewys dat indien $f'(x) > 0$ op 'n interval I , dan is f stygend op die interval I .

Let x_1, x_2 be any numbers in I such that $x_1 < x_2$
Then

f is diff on (x_1, x_2)

f is cont on $[x_1, x_2]$



From the Mean Value Theorem
it follows that there exists a
 $c \in (x_1, x_2)$ such that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

But $f'(c) > 0$ and $x_2 - x_1 > 0$

So $f(x_2) - f(x_1) > 0$

$$\Rightarrow f(x_2) > f(x_1)$$

3

Question 5 / Vraag 5

Given that the function $f(x) = x \ln x$ has an axis-intercept $(1, 0)$ and no asymptotes. Also given that $\lim_{x \rightarrow 0^+} f(x) = 0$. Sketch the function on $(0, 2]$ by making use of the first and the second derivatives and the necessary tests. Indicate all extremes and show all calculations.

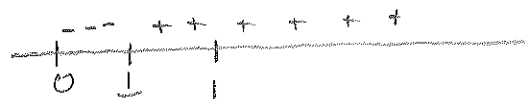
Gegee dat die funksie $f(x) = x \ln x$ 'n asse-snyding by $(1, 0)$ het en geen asimptote nie. Ook gegee dat $\lim_{x \rightarrow 0^+} f(x) = 0$. Skets die funksie op $(0, 2]$ deur gebruik te maak van die eerste en tweede afgeleides en die nodige toetse. Dui alle ekstreme aan en toon alle berekeninge.

Hint / Wenk: $\frac{1}{e} \approx 0.37$ and / en $\ln 2 \approx 0.7$

$$f'(x) = \ln x + x \cdot \frac{1}{x} = \ln x + 1$$

Critical number: $\ln x + 1 = 0 \Rightarrow x = e^{-1} \approx 0.37$

First derivative test:



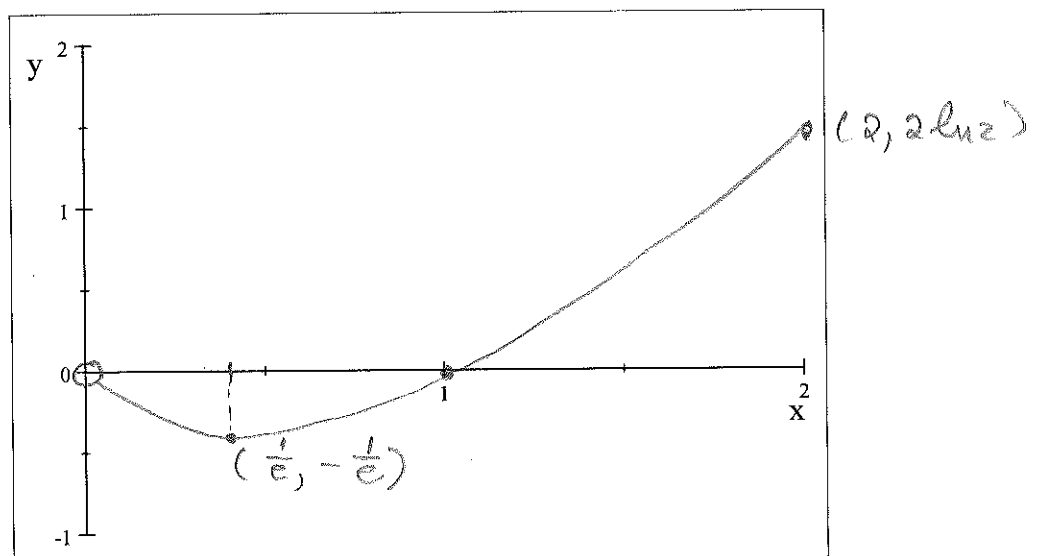
$\Rightarrow f$ has a local minimum in $x = e^{-1}$ with $f(e^{-1}) = -\frac{1}{e} \approx -0.37$

Second derivative:

$$f''(x) = \frac{1}{x} \therefore \text{So } f''(x) > 0 \text{ for all } x > 0$$

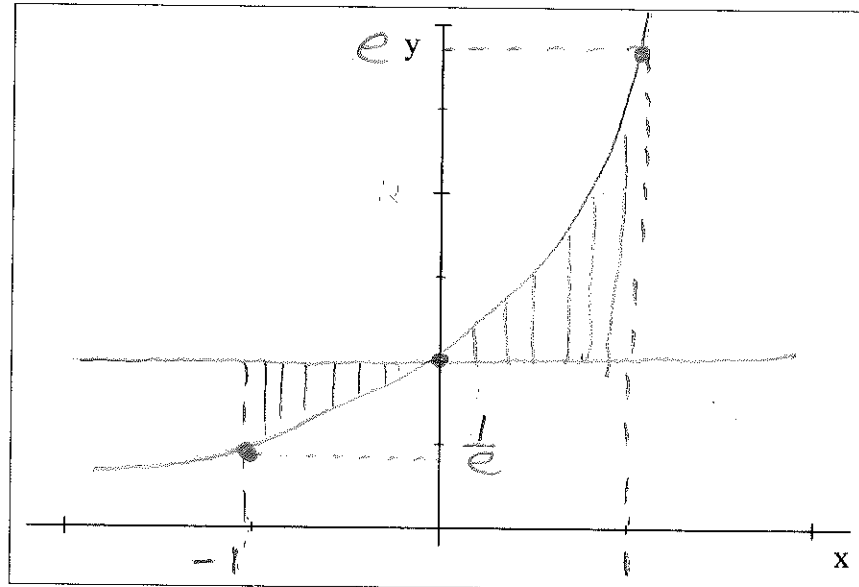
$$\Rightarrow f \text{ is concave up for } x > 0$$

Also: $f(2) = 2 \ln 2 \approx 1.4$



Question 6 / Vraag 6

6.1 Sketch the curves $y = e^x$ and $y = 1$ on the same set of axes on the interval $[-1, 1]$.
 Skets die krommes $y = e^x$ en $y = 1$ op dieselfde assestelsel op die interval $[-1, 1]$.



1

6.2 Find the area enclosed between the curve $y = e^x$ and $y = 1$ on $[-1, 1]$.
 Bepaal die oppervlakte ingesluit tussen die krommes $y = e^x$ en $y = 1$ op $[-1, 1]$.

$$\begin{aligned}
 \text{Area: } & 1 - \int_{-1}^0 e^x \cdot dx + \int_0^1 e^x dx - 1 \\
 & = -e^x \Big|_{-1}^0 + e^x \Big|_0^1 \\
 & = -(1 - e^{-1}) + e^1 - 1 \\
 & = e^1 + e^{-1} - 2
 \end{aligned}$$

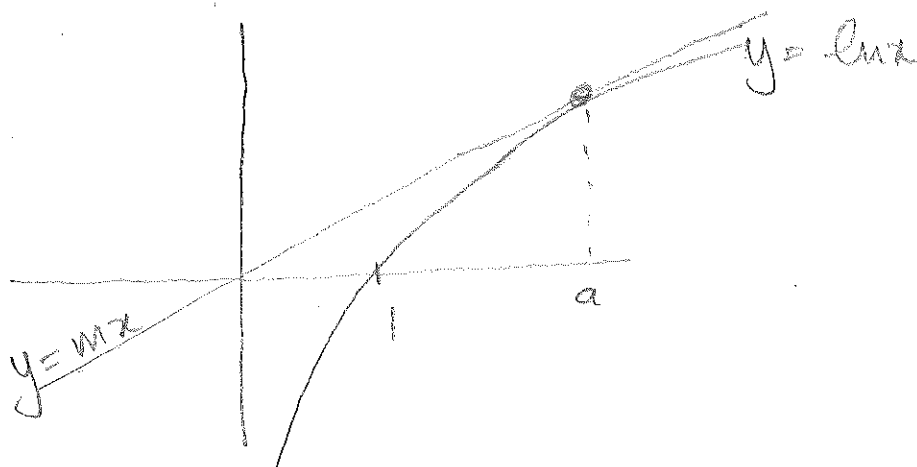
3

$$\begin{aligned}
 & \int_{-1}^0 (1 - e^x) dx + \int_0^1 (e^x - 1) dx \\
 & = (x - e^x) \Big|_{-1}^0 + (e^x - x) \Big|_0^1 \\
 & = -1 - (-1 - e^{-1}) + e^1 - 1 - 1 \\
 & = e^{-1} + e^1 - 2
 \end{aligned}$$

Question 7 / Vraag 7

Find the equation of a line through the origin that is tangent to the graph of $y = \ln x$.

Vind die vergelyking van die lyn deur die oorsprong wat 'n raaklyn is aan die grafiek van $y = \ln x$.



Derivatives at are the same at tangent point:

$$\frac{1}{a} = m$$

Function values are the same at tangent point:

$$ma = \ln a$$

$$\Rightarrow 1 = \ln a \Rightarrow a = e$$

$$\Rightarrow y - 0 = \frac{1}{e}(x - 0)$$

$$\Rightarrow y = \frac{1}{e}x = e^{-1}x$$

OR: Tangent line is $y - \ln a = \frac{1}{a}(x - a)$

The line passes through $(0,0)$

$$\Rightarrow -\ln a = -1$$

$$\Rightarrow \ln a = 1$$

$$\Rightarrow a = e$$

So the equation of the line is $y = \frac{1}{e}x$.